

Prime Number Assembly Project

[Click this link](#)

To go to the whole project on GitHub

[Click this link](#)

To go to Prime Number Checker Assembly Project (C++) file link

[Click this link](#)

To go to Prime Number Checker Assembly Project (Python) file link

Memory Segment Size Summary

Segment	Start Address	End Address	Size (Hex)	Size (Decimal)
.text	0x00400000	0x004001CC	0x1D0 bytes	464 bytes
.data	0x10010000	0x10010106	0x106 bytes	262 bytes

Assembly Mips Text Segment

Address	Code	Source	Label
0x00400000	0x24020004	li \$v0, 4	main
0x00400004	0x3c011001	la \$a0, header_info	
0x00400008	0x3424002b		
0x0040000c	0x0000000c	syscall	
0x00400010	0x24020004	li \$v0, 4	
0x00400014	0x3c011001	la \$a0, lbl_name	
0x00400018	0x34240049		
0x0040001c	0x0000000c	syscall	
0x00400020	0x24020004	li \$v0, 4	
0x00400024	0x3c011001	la \$a0, lbl_section	
0x00400028	0x34240073		
0x0040002c	0x0000000c	syscall	

0x00400030	0x24020004	li \$v0, 4	
0x00400034	0x3c011001	la \$a0, header_init	
0x00400038	0x34240081		
0x0040003c	0x0000000c	syscall	
0x00400040	0x2402000b	li \$v0, 11	
0x00400044	0x2404004d	li \$a0, 'M'	
0x00400048	0x0000000c	syscall	
0x0040004c	0x24020004	li \$v0, 4	
0x00400050	0x3c011001	la \$a0, separator	
0x00400054	0x342400a1		
0x00400058	0x0000000c	syscall	
0x0040005c	0x24020001	li \$v0, 1	
0x00400060	0x2404004d	li \$a0, 77	
0x00400064	0x0000000c	syscall	
0x00400068	0x24020004	li \$v0, 4	
0x0040006c	0x3c011001	la \$a0, newline	
0x00400070	0x342400a5		
0x00400074	0x0000000c	syscall	

0x00400078	0x2402000b	li \$v0, 11	
0x0040007c	0x24040054	li \$a0, 'T'	
0x00400080	0x0000000c	syscall	
0x00400084	0x24020004	li \$v0, 4	
0x00400088	0x3c011001	la \$a0, separator	
0x0040008c	0x342400a1		
0x00400090	0x0000000c	syscall	
0x00400094	0x24020001	li \$v0, 1	
0x00400098	0x24040054	li \$a0, 84	
0x0040009c	0x0000000c	syscall	
0x004000a0	0x24020004	li \$v0, 4	
0x004000a4	0x3c011001	la \$a0, newline	
0x004000a8	0x342400a5		
0x004000ac	0x0000000c	syscall	
0x004000b0	0x2402000b	li \$v0, 11	
0x004000b4	0x2404004d	li \$a0, 'M'	
0x004000b8	0x0000000c	syscall	
0x004000bc	0x24020004	li \$v0, 4	

0x004000c0	0x3c011001	la \$a0, separator	
0x004000c4	0x342400a1		
0x004000c8	0x0000000c	syscall	
0x004000cc	0x24020001	li \$v0, 1	
0x004000d0	0x2404004d	li \$a0, 77	
0x004000d4	0x0000000c	syscall	
0x004000d8	0x24020004	li \$v0, 4	
0x004000dc	0x3c011001	la \$a0, newline	
0x004000e0	0x342400a5		
0x004000e4	0x0000000c	syscall	
0x004000e8	0x2402000b	li \$v0, 11	
0x004000ec	0x24040045	li \$a0, 'E'	
0x004000f0	0x0000000c	syscall	
0x004000f4	0x24020004	li \$v0, 4	
0x004000f8	0x3c011001	la \$a0, separator	
0x004000fc	0x342400a1		
0x00400100	0x0000000c	syscall	
0x00400104	0x24020001	li \$v0, 1	

0x00400108	0x24040045	li \$a0, 69	
0x0040010c	0x0000000c	syscall	
0x00400110	0x24020004	li \$v0, 4	
0x00400114	0x3c011001	la \$a0, newline	
0x00400118	0x342400a5		
0x0040011c	0x0000000c	syscall	
0x00400120	0x24020004	li \$v0, 4	
0x00400124	0x3c011001	la \$a0, divider	
0x00400128	0x342400a7		
0x0040012c	0x0000000c	syscall	
0x00400130	0x24020004	li \$v0, 4	
0x00400134	0x3c011001	la \$a0, prompt_msg	
0x00400138	0x342400c4		
0x0040013c	0x0000000c	syscall	
0x00400140	0x24020005	li \$v0, 5	
0x00400144	0x0000000c	syscall	
0x00400148	0x00024021	move \$t0, \$v0	
0x0040014c	0x24090001	li \$t1, 1	

0x00400150	0x0128082a	ble \$t0, \$t1, not_prime	
0x00400154	0x10200015		
0x00400158	0x24090002	li \$t1, 2	
0x0040015c	0x1109000b	beq \$t0, \$t1, is_prime	
0x00400160	0x31090001	andi \$t1, \$t0, 1	
0x00400164	0x11200011	beqz \$t1, not_prime	
0x00400168	0x240a0003	li \$t2, 3	
0x0040016c	0x00085842	srl \$t3, \$t0, 1	
0x00400170	0x016a082a	bgt \$t2, \$t3, is_prime	check_loop
0x00400174	0x14200005		
0x00400178	0x010a001a	div \$t0, \$t2	
0x0040017c	0x00006010	mfhi \$t4	
0x00400180	0x1180000a	beqz \$t4, not_prime	
0x00400184	0x214a0002	addi \$t2, \$t2, 2	
0x00400188	0x0810005c	j check_loop	
0x0040018c	0x24020001	li \$v0, 1	is_prime
0x00400190	0x00082021	move \$a0, \$t0	
0x00400194	0x0000000c	syscall	

0x00400198	0x24020004	li \$v0, 4	
0x0040019c	0x3c011001	la \$a0, prime_msg	
0x004001a0	0x342400d8		
0x004001a4	0x0000000c	syscall	
0x004001a8	0x08100072	j exit	
0x004001ac	0x24020001	li \$v0, 1	not_prime
0x004001b0	0x00082021	move \$a0, \$t0	
0x004001b4	0x0000000c	syscall	
0x004001b8	0x24020004	li \$v0, 4	
0x004001bc	0x3c011001	la \$a0, not_prime_msg	
0x004001c0	0x342400ed		
0x004001c4	0x0000000c	syscall	
0x004001c8	0x2402000a	li \$v0, 10	exit
0x004001cc	0x0000000c	syscall	

Assembly Mips Labels

Label	Address
main	0x00400000
check_loop	0x00400170
is_prime	0x0040018c
not_prime	0x004001ac
exit	0x004001c8
full_name	0x10010000
section_num	0x1001001f
header_info	0x1001002b
lbl_name	0x10010049
lbl_section	0x10010073
header_init	0x10010081
separator	0x100100a1
newline	0x100100a5
divider	0x100100a7
prompt_msg	0x100100c4
prime_msg	0x100100d8
not_prime_msg	0x100100ed